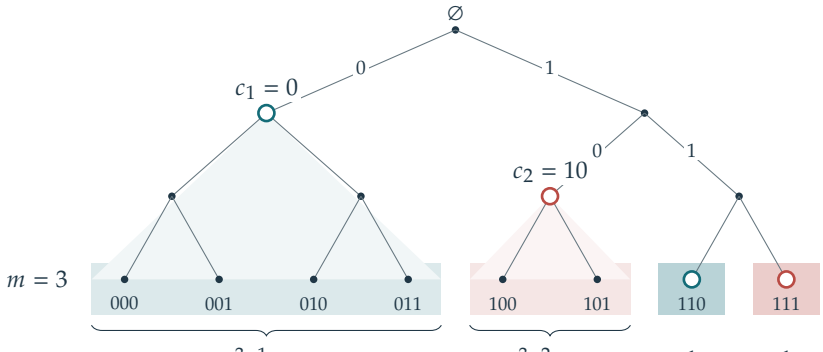
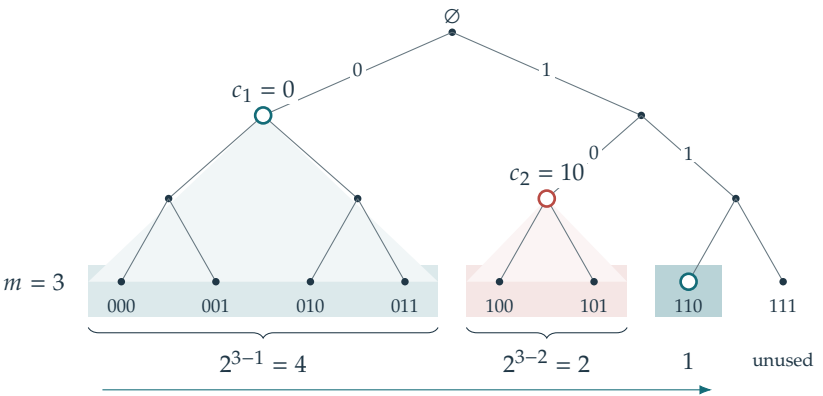
Panel 1



A codeword of length ℓ reserves $2^{m-\ell}$ leaves at depth m.

$$c = (0, 10, 110, 111), \qquad \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = 1$$

Panel 2



$$\ell = (1, 2, 3), \quad c = (0, 10, 110), \quad \textstyle\sum_i 2^{-\ell_i} = \frac{7}{8} \leq 1$$